

Erik H. Kramer, Ph.D.

Skills

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Technical	Mechanical design, Systems integration, GD&T, FEA, Test design, V&V, Failure analysis, Data analysis, Rapid prototyping, Robot control
Hardware	Robotic arms, Mobility systems, 3D Printers, Sensors, Material test & qualification systems, Soldering, Wire bonder, Machine shop tools
Software	SolidWorks, Teamcenter, Linux, Git, ROS, Bluehill Universal, Vicon Tracker, Slicing software, Photoshop, Excel, Google Workspace
Coding	C/C++, MATLAB, Python, LaTeX

Robotics Hardware & Mechanical Systems Experience

NASA Jet Propulsion Laboratory - ROBOTICS MECHANICAL ENGINEER III

July 2026 - Present

- Owned dynamometer development for flight actuator verification, designing hardware, harnessing, software, and executing ongoing load testing
- Responsible engineer for two mobile manipulation robotic testbeds, developing and maintaining electrical, mechanical, and software systems
- Leading development and delivery of next-gen rover steering actuators, performing mobility & load-case analysis and electromechanical design

POSTDOCTORAL FELLOW

Mar 2025 - July 2026

- Led technical execution for TRL-4/5 autonomous system demos of flight-like satellite servicing, rover payload transfer, and telescope assembly
- Served as Deputy Task Manager for a 6-person lunar solar array assembly technology development team, leading technical efforts to meet milestones
- Diagnosed, repaired, and documented root causes of actuator failures through hardware teardown and reassembly, restoring system functionality
- Operated multiple robotic arm platforms for test campaigns, rapidly resolving electromechanical and controls issues in real time to maintain progress
- Developed driving algorithms for a robotic mobility system and analyzed field-test data to validate algorithm functionality and actuator performance
- Designed robotic sampling sequences to validate lunar scoop collection of regolith and rocks over repeated passes for a mission concept review

ROBOTICS INTERN

Mar 2022 - Dec 2024

- Developed and delivered the central structure for a hazardous environment rover, performing design, analysis, procurement, and integration
- Designed a new robotic arm optimized for sampling kinematics, meeting torque and stiffness requirements while leveraging heritage hardware
- Applied rapid design techniques to quickly develop test rigs and robot hardware components while supporting rover field testing campaigns
- Architected features for a robot arm & lander testbed using C++, such as a torque emulation algorithm and a ROS interface for an autonomy subsystem
- Optimized motion planning for a redundant 7-DOF robotic arm using analytic inverse kinematics, increasing algorithm speed by 30%
- Analyzed large datasets of robot telemetry to understand the root cause of unexpected hardware faults during operation and implement code fixes

UCLA Bionics Lab - GRADUATE RESEARCH ENGINEER

Sept 2015 - Dec 2024

- Led a team in the end-to-end development of a robotic exoskeleton, providing hands-on work for design, fabrication, assembly, testing, and software
- Integrated force-torque sensors with 5 industrial robotic arms by developing real-time C++ software, resulting in safe multi-robot coordinated control
- Built MuJoCo dynamics simulations and admittance control algorithms, enabling safe human-robot haptic interaction with virtual environments
- Translated high-level requirements into hardware configurations through kinematic workspace analysis, achieving >96% coverage of potential users
- Collaborated with analysts to perform hand calculations and conduct stress FEA, ensuring safety factors of structural weldments met requirements
- Utilized CAD to design small connectors and large structures for manufacturability with traditional processes such as CNC and welding

Additional Engineering & Science Experience

Physics & Astronomy Department at UCLA - LEAD TEACHING FELLOW

Sept 2016 - Mar 2022

- Managed multiple small teams teaching space science, circuit design, microcontrollers, E&M, Python, data analysis, technical writing, and mechanics

Super Cryogenic Dark Matter Search Collaboration - UNDERGRADUATE RESEARCHER

Feb 2013 - June 2015

- Designed metallic and composite cryogenic instrument thermal standoffs and drafted part drawings using ASME Y14.5 GD&T standards
- Conducted stress and thermal simulations via FEA and custom MATLAB scripts on 3 potential designs to support trade studies
- Delivered mechanical test support equipment and used it to qualify standoff designs through strength and cryogenic thermal V&V testing

Berkeley Particle Cosmology Group - UNDERGRADUATE RESEARCHER

Feb 2013 - June 2015

- Translated rough ideas from scientific experts into functional mechanical hardware, such as cold plates, for sub-kelvin high-vacuum environments
- Assembled hardware in a cleanroom and maintained low-temp experiments in bath cryostat vacuum chambers and a dilution refrigerator
- Created procedures and documented build and test instructions for experiments determining low-temp properties of engineering materials

Independent Projects & Mentorship

FIRST Robotics Mentor - TEAM 1138

Jan 2026 - Present

- Guiding high school students through mechanical design, electrical integration, software development, and design reviews for competition robots

Costume & Mechatronic Prop Design

July 2006 - Present

- Built replicas and wearables, including a Mars 2020 RC rover and an animatronic tail by utilizing mechanical, electrical, and software design processes

Education

University of California, Los Angeles (UCLA) - PH.D. AND M.S. IN MECHANICAL ENGINEERING

- Dissertation: *Robotic Physical Emulation of Virtual Dynamic Environments: Applications in Full Body Haptics and Spacecraft Development*

University of California, Berkeley - B.A. IN PHYSICS AND B.S. IN MECHANICAL ENGINEERING